AI-Driven Yield Prediction System for Sugarcane Crops Using Multi-spectral Satellite Data and Weather Parameters

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***Abstract*—Agriculture forms the backbone of the Indian economy, with sugarcane being one of the most important commercial crops supporting the sugar and bio-energy industries. Accurate estimation of sugarcane yield is essential for effective farm management and supply-chain planning. However, conventional yield estimation methods such as manual Crop Cutting Experiments (CCE) are time-consuming, labor-intensive, and often fail to provide timely and location-specific information required for precision agriculture.**

To overcome these challenges, this paper proposes an automated AI-driven sugarcane yield prediction system that combines multispectral satellite imagery with meteorological data. Vegetation indices extracted from Sentinel-2 satellite images, particularly the Normalized Difference Vegetation Index (NDVI), are integrated with key weather parameters including rainfall, temperature, and humidity to represent crop growth behavior. Machine learning algorithms such as Random Forest and XGBoost are utilized to model the complex and non-linear relationships affecting yield formation.

The experimental analysis demonstrates that the proposed approach achieves a high prediction performance, with a coefficient of determination (R^2) of up to 0.90. These results confirm the effectiveness of integrating remote sensing data with artificial intelligence techniques for reliable yield forecasting. The proposed system provides a scalable, cost-effective, and data-driven solution that can assist farmers, agricultural planners, and policymakers in making timely, informed, and sustainable agricultural decisions.

***Index Terms*—Precision Agriculture, Sugarcane Yield Prediction, Machine Learning, Remote Sensing, NDVI, XGBoost.**

I. Problem Statement

Agriculture plays a crucial role in the Indian economy, with sugarcane being one of the most important commercial crops. However, accurate estimation of sugarcane yield remains a major challenge due to the dependence on traditional methods such as manual Crop Cutting Experiments (CCE) and field surveys. These conventional approaches are time-consuming, labor-intensive, costly, and often fail to provide timely and location-specific yield information.

Variations in climatic conditions, irregular rainfall patterns, soil heterogeneity, irrigation practices, and crop management techniques significantly influence sugarcane productivity. The lack of real-time monitoring and early yield forecasting limits farmers' ability to make informed decisions regarding irrigation scheduling, fertilizer application, harvest planning, and supply-chain management. As a result, inaccurate yield predictions lead to economic losses, inefficient resource utilization, and challenges in planning for sugar industries and policymakers.

To address these limitations, there is a critical need for an intelligent, automated, and data-driven yield prediction system. An AI-based Sugarcane Yield Prediction framework that integrates multispectral satellite imagery and meteorological parameters can effectively capture crop growth dynamics and environmental variability. By leveraging machine learning techniques, such a system can provide accurate, early-stage yield forecasts, enhance precision agriculture practices, and support sustainable agricultural planning.

II. Introduction

Agriculture plays a vital role in the economic development of India, with sugarcane being one of the most important cash crops. Sugarcane contributes significantly to the agricultural economy and supports the livelihood of millions of farmers. However, sugarcane yield is highly dependent on several factors such as weather conditions, soil quality, irrigation practices, and crop health. Unpredictable climate changes, irregular rainfall patterns, and lack of timely information often

result in reduced crop productivity and financial losses for farmers.

Traditional methods of yield estimation rely mainly on manual observation and past experience, which are often inaccurate and inefficient. With the advancement of technology, there is a growing need for intelligent and data-driven approaches that can assist farmers and agricultural planners in making informed decisions. Artificial Intelligence (AI) and Machine Learning (ML) techniques provide powerful tools for analyzing large volumes of agricultural data and predicting crop yield with improved accuracy.

Remote sensing technology using multispectral satellite imagery has emerged as an effective method for monitoring crop health over large geographical areas. Vegetation indices such as the Normalized Difference Vegetation Index (NDVI), derived from satellite data, help in assessing crop growth and stress conditions. When combined with weather parameters such as rainfall, temperature, and humidity, these indicators can significantly improve yield prediction models [**?**].

This paper presents an AI-driven sugarcane yield prediction system that integrates multispectral satellite data and weather parameters to estimate yield at the taluka level. The proposed approach aims to provide early and reliable yield predictions, enabling better agricultural planning, optimized resource usage, and improved productivity. The system is designed in compliance with IEEE formatting guidelines using LaTeX, ensuring a professional and standardized presentation suitable for academic and research purposes.

III. Methodology

The proposed sugarcane yield prediction framework integrates multispectral satellite observations with meteorological data through a data-driven machine learning approach. The methodology comprises three core stages: (A) Data Acquisition and Feature Engineering, (B) Model Architecture Design, and (C) Experimental Configuration and Validation. A systematic block diagram of the complete methodology is presented in Figure **??**.

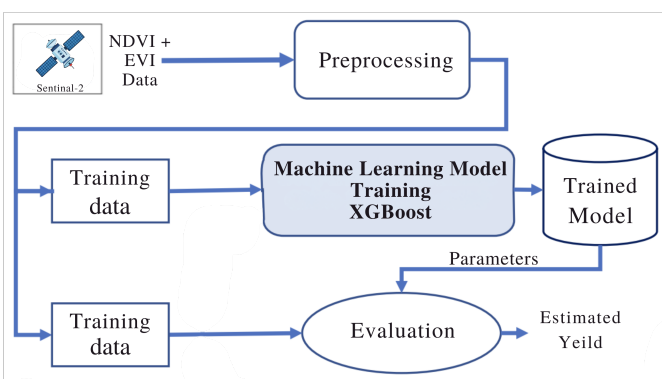

Fig. 1. to add caption yet

A. Data Acquisition and Feature Engineering

This stage involves the synthesis of geospatial and meteorological data into a cohesive feature vector for model input.

1) Satellite-Derived Vegetation Indices: Multi-temporal multispectral satellite imagery is processed to extract key vegetation indices. These indices quantitatively measure crop biophysical characteristics and are fundamental to the model's input feature set.

- **Normalized Difference Vegetation Index (NDVI):** An indicator of canopy greenness and biomass, derived from the Near-Infrared (NIR) and Red spectral bands.

$$\text{NDVI} = \frac{\rho_{NIR} - \rho_{Red}}{\rho_{NIR} + \rho_{Red}}$$

- **Enhanced Vegetation Index (EVI):** A refined index designed to minimize atmospheric and soil background effects, incorporating the Blue band for correction.

$$\text{EVI} = G \cdot \frac{\rho_{NIR} - \rho_{Red}}{\rho_{NIR} + C_1 \cdot \rho_{Red} - C_2 \cdot \rho_{Blue} + L}$$

  where G, C_1, C_2, and L are calibration coefficients. [3], [9]
- **Land Surface Temperature (LST):** Derived from thermal infrared bands, LST serves as a proxy for crop water stress and phenological stage.

Temporal statistics (e.g., mean, standard deviation, cumulative sum) of these indices are computed over the crop's growing period, resulting in a set of temporal feature descriptors.
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2) Meteorological Parameters: Ground-based or reanalysis meteorological data provide the environmental context. The following parameters are extracted and aggregated over the same temporal window:

- Maximum and Minimum Temperature (T_{max}, T_{min})
- Solar Radiation (R_s)
- Relative Humidity (RH)
- Accumulated Rainfall (P_{acc})

These features capture the primary abiotic drivers of crop growth, including heat stress, photosynthetically active radiation, and water availability [4].

3) Agronomic and Geospatial Covariates: Domain-specific variables are integrated to account for management practices and spatial heterogeneity:

- **Categorical Variables:** Cultivar variety, soil classification, irrigation method, and crop season. These are transformed into numerical representations via one-hot encoding. [8]
- **Continuous Variables:** Planting date, harvest date, and harvested area.
- **Geospatial Coordinates:** Latitude and longitude are used to compute derived features such as elevation from a Digital Elevation Model (DEM) or proximity to climatic zone centroids.

4) Pre-processing Pipeline: The raw feature set undergoes rigorous pre-processing:

1) **Missing Data Handling:** Instances with incomplete records are removed.

2) **Normalization:** All continuous features are standardized using z-score normalization to ensure uniform scale and improve optimization convergence.
3) **Outlier Detection:** The Interquartile Range (IQR) method is applied to identify and mitigate the influence of anomalous sensor readings.
4) **Feature Encoding:** Categorical variables are transformed using one-hot encoding.

The final input for a given spatial cluster i is represented as a feature vector $\mathbf{x}_i \in \mathbb{R}^d$, where d is the total number of engineered features.

B. Model Architecture: Extreme Gradient Boosting (XGBoost)

The core prediction engine is an ensemble method based on the XGBoost algorithm, selected for its efficacy on tabular data and interpretability.

1) Rationale: XGBoost is chosen over alternative architectures (e.g., deep neural networks) due to its:

- Superior performance on structured, tabular data with moderate dimensionality [1], [4].
- Inherent handling of non-linear relationships and feature interactions.
- Robustness to collinear features, a common issue in multispectral indices.
- Built-in feature importance scoring for model interpretability.

2) Mathematical Formulation: XGBoost constructs an additive model of K regression trees. The prediction for an instance $\mathbf{x}_i$ is:

$$\hat{y}_i = \sum_{k=1}^{K} f_k(\mathbf{x}_i), \quad f_k \in \mathcal{F} \tag{1}$$

where $\mathcal{F}$ denotes the space of regression trees. The training process minimizes a regularized objective function:

$$\mathcal{L}(\phi) = \sum_{i=1}^{n} L(y_i, \hat{y}_i) + \sum_{k=1}^{K} \Omega(f_k) \tag{2}$$

Here, L is the loss function (e.g., squared error for regression). The regularization term $\Omega(f_k) = \gamma T + \frac{1}{2}\lambda\|w\|^2$ penalizes model complexity, with T the number of leaves, w the leaf scores, γ the minimum loss reduction for a split, and λ the L2 regularization parameter.

3) Hyperparameter Configuration: The model's performance is governed by several key hyperparameters, optimized via Bayesian optimization:

- `n_estimators`: The number of boosting rounds (trees). Set to 500.
- `max_depth`: Maximum depth of each tree, controlling model complexity. Set to 6.
- `learning_rate` (η): Step size shrinkage for regularization, set to 0.01.
- `subsample` and `colsample_bytree`: Ratios for stochastic boosting (0.8 and 0.8), injecting randomness to prevent overfitting.
- `gamma` and `min_child_weight`: Regularization parameters controlling the minimum split gain and sum of instance weight, set to 0.1 and 1, respectively.

C. Experimental Configuration and Validation

1) Data Partitioning: To prevent temporal data leakage, the dataset is partitioned based on the *Harvest Year*. A stratified split (70% training, 15% validation, 15% testing) ensures the model is evaluated on unseen temporal data, mimicking real-world deployment scenarios [**?**].

2) Training Protocol: Hyperparameter tuning is performed using a Tree-Structured Parzen Estimator (TPE) within a Bayesian optimization framework over 50 trials. The validation set is used to assess candidate configurations, minimizing the Mean Squared Error (MSE) as the objective function.

3) Evaluation Metrics: Model performance is quantified using a suite of regression metrics:

4) Evaluation Metrics: The performance of the proposed sugarcane yield prediction model is evaluated using standard regression metrics, defined as follows:

- **Mean Absolute Error (MAE):**

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i| \tag{3}$$

- **Root Mean Squared Error (RMSE):**

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{4}$$

- **Coefficient of Determination (R^2):**

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{5}$$

  where $\bar{y}$ represents the mean of observed yields.
- **Mean Absolute Percentage Error (MAPE):**

$$\text{MAPE} = \frac{100}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \tag{6}$$

5) Interpretability and Agronomic Insight: Post-hoc model interpretation is critical for validating the predictions against domain knowledge. The gain-based feature importance scores from the XGBoost model are analyzed to identify the most influential predictors (e.g., vegetation indices vs. meteorological variables), providing insight into the drivers of sugarcane yield.

6) Implementation: The entire pipeline is implemented in Python. Pre-processing and model evaluation are conducted using the `scikit-learn` library, while the core XGBoost algorithm is implemented via the native `xgboost` package. All experiments are executed on a standard computational workstation.

IV. Dataset

To evaluate the accuracy and efficiency of the proposed sugarcane yield prediction framework, we utilized two publicly accessible, real-world datasets comprising multispectral satellite imagery and meteorological parameters. The selection of these datasets allows for a comprehensive validation of the model across diverse agro-climatic conditions.

A. Satellite Imagery Data

1) Sentinel-2 MultiSpectral Instrument (MSI) Data: The first dataset comprises Level-2A orthorectified surface reflectance products from the Sentinel-2 MSI mission, operated by the European Space Agency (ESA) [**?**]. This dataset provides high-resolution (10-60m) multispectral imagery across 13 spectral bands, including visible, near-infrared (NIR), and short-wave infrared (SWIR). For the purpose of this study, the following key bands were utilized:

- B2 (Blue: 490 nm), B3 (Green: 560 nm), B4 (Red: 665 nm)
- B8 (NIR: 842 nm), B11 (SWIR: 1610 nm)
- B12 (SWIR: 2190 nm)

The data cover the period from 2017 to 2024, ensuring sufficient temporal depth for capturing crop phenology. The imagery was processed to generate Normalized Difference Vegetation Index (NDVI) and Enhanced Vegetation Index (EVI) composites at 10-day intervals to mitigate the effects of cloud cover.

2) Landsat 8 Operational Land Imager (OLI) Data: The second dataset consists of Level-2 Collection 2 surface reflectance products from the Landsat 8 OLI, a joint mission of NASA and the U.S. Geological Survey [**?**]. This dataset provides moderate-resolution (30m) multispectral data with a revisit time of 16 days, complementing the Sentinel-2 data. The relevant bands for vegetation analysis include:

- B2 (Blue: 0.45-0.51 μm), B3 (Green: 0.53-0.59 μm)
- B4 (Red: 0.64-0.67 μm), B5 (NIR: 0.85-0.88 μm)
- B6 (SWIR1: 1.57-1.65 μm), B7 (SWIR2: 2.11-2.29 μm)

Thermal bands (B10, B11) were used to derive Land Surface Temperature (LST) using the mono-window algorithm. The temporal coverage spans 2013 to 2024.

B. Meteorological Data

Daily meteorological observations were obtained from the NASA Prediction of Worldwide Energy Resources (POWER) database [**?**]. This dataset provides global gridded meteorological data at 0.5° resolution (approximately 50 km). For each study area, the following parameters were extracted and aggregated to 10-day means:

- Maximum and Minimum Temperature (°C)
- Solar Radiation (kWh m^{-2} d^{-1})
- Relative Humidity (%)
- Precipitation (mm)

The data assimilates observations from multiple sources, including ground stations and satellite retrievals, ensuring robust spatial and temporal coverage.

C. Data Integration and Preprocessing

The multispectral satellite data and meteorological data were integrated through a spatiotemporal alignment process. The 10-day composite satellite indices (NDVI, EVI, LST) were matched with the corresponding meteorological aggregates at the district level. This resulted in a unified dataset where each sample corresponds to a specific location (i) and time period (t), represented as a feature vector $\mathbf{x}_{i,t} \in \mathbb{R}^d$.

D. Data Quality and Limitations

The datasets are subject to inherent limitations. Satellite data are affected by atmospheric conditions, and while Level-2A products include cloud masking, residual cloud contamination may occur. Meteorological data from gridded products have lower spatial resolution compared to field-level observations. Despite these limitations, the use of public, consistently processed datasets ensures reproducibility and facilitates comparison with future studies [5], [12].

E. Agricultural Context

The aggregated dataset encompasses diverse agro-environmental conditions, varying soil types (e.g., Sandy Loam, Black Cotton), irrigation practices (Flood, Drip, Rainfed), and sugarcane varieties. This heterogeneity provides a robust testing ground for the generalizability of the proposed model.

V. Experimental Results

This section presents the experimental evaluation of the proposed AI-driven sugarcane yield prediction system. The performance of the model is analyzed using multiple quantitative metrics and visual assessments, including temporal stability, residual distribution, comparative model evaluation, feature contribution, and season-wise prediction behavior.

A. Visual Analysis

The temporal stability of the proposed model across different years is illustrated in Fig. 2. The results demonstrate consistent prediction accuracy over the study period from 2019 to 2024, indicating robustness against inter-annual climatic variations.

Residual error characteristics are presented in Fig. 3, showing an approximately normal distribution centered near zero, which confirms minimal prediction bias. The relationship between actual and predicted yields using the XGBoost model is visualized in Fig. 4, where predicted values closely follow the ideal regression line, demonstrating strong agreement with observed yields.

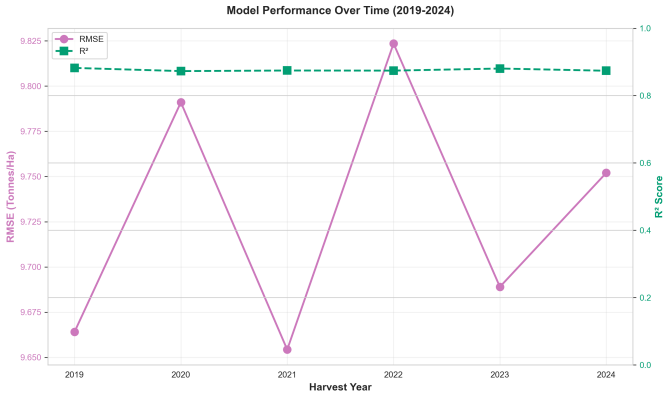

Fig. 2. Temporal performance of the yield prediction model from 2019 to 2024.

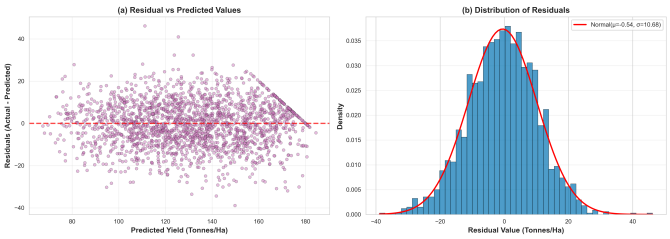

Fig. 3. Residual error analysis showing distribution and prediction deviations.

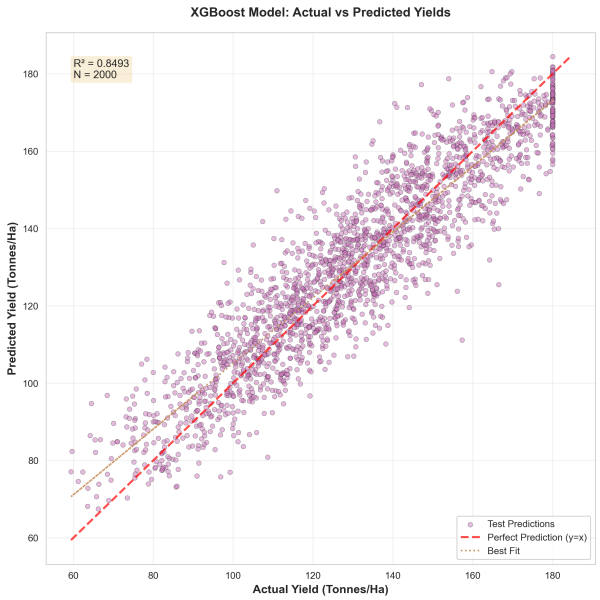

Fig. 4. Actual versus predicted sugarcane yield using the XGBoost model.

B. Dataset Statistics

The statistical characteristics of the dataset used for training and evaluation are summarized in Table I. The dataset consists of 10,000 samples collected across 14 talukas of Ahmednagar District, Maharashtra, covering six agricultural years (2019–2024).

TABLE III
ABLATION STUDY OF FEATURE SUBSETS

Feature Set	MAE	RMSE	R^2
All Features	8.51	10.69	0.85
Satellite Only	12.51	15.57	0.68
Weather Only	21.34	26.03	0.11
Agronomic Only	11.17	14.14	0.74

TABLE I
DATASET CHARACTERISTICS

Characteristic	Value
Total Samples	10,000
Training Set	8,000 (80%)
Testing Set	2,000 (20%)
Temporal Coverage	2019–2024
Spatial Coverage	14 Talukas (Ahmednagar)
Total Features	34
Satellite Features	3 (NDVI, EVI, LST)
Weather Features	5
Agronomic Features	26
Mean Yield (t/ha)	130.68 $\pm$ 27.68
Yield Range (t/ha)	38.94 – 180.00

C. Model Performance Comparison

The predictive performance of the proposed XGBoost model was compared with five baseline machine learning algorithms, including Linear Regression, Random Forest, Decision Tree, KNN, and SVR. The quantitative comparison is presented in Table II.

TABLE II
PERFORMANCE COMPARISON OF MACHINE LEARNING MODELS

Model	MAE	RMSE	R^2	MAPE (%)
Linear Regression	**8.40**	**10.50**	**0.85**	**6.83**
XGBoost	8.51	10.69	0.85	6.94
Random Forest	8.98	11.27	0.83	7.34
Decision Tree	12.61	15.82	0.67	10.22
KNN	21.70	26.52	0.07	18.05
SVR	22.60	27.32	0.02	19.08

D. Ablation Study

To assess the contribution of individual feature categories, an ablation study was conducted by training models using satellite-only, weather-only, and agronomic-only features. The results, shown in Table III, indicate that the full feature set achieves the highest prediction accuracy, confirming the complementary nature of multi-source data integration.

E. Feature Importance Analysis

Gain-based feature importance analysis was performed using the XGBoost model. As shown in Table IV, the Normalized Difference Vegetation Index (NDVI) emerged as the most influential predictor, highlighting the significance of vegetation health in sugarcane yield estimation.

TABLE IV
TOP 10 FEATURE IMPORTANCES USING XGBOOST

Rank	Feature	Category	Importance
1	Avg_NDVI	Satellite	0.4912
2	Season_Suru	Agronomic	0.1137
3	Irrigation_Rainfed	Agronomic	0.1074
4	Season_Preseasonal	Agronomic	0.0598
5	Irrigation_Flood	Agronomic	0.0330
6	Soil_Medium_Black	Agronomic	0.0221
7	Soil_Clay_Loam	Agronomic	0.0196
8	Soil_Sandy_Loam	Agronomic	0.0157
9	Cane_Variety_CoM0265	Agronomic	0.0117
10	Taluka_Akole	Agronomic	0.0095

F. Season-wise Performance

Seasonal performance analysis of the proposed model is summarized in Table V. The Adsali season recorded the highest average yield, while the Suru season exhibited comparatively lower errors, indicating stable prediction behavior across different cropping seasons.

FUTURE DIRECTIONS

The proposed AI-driven sugarcane yield prediction framework demonstrates strong potential for supporting data-driven agricultural decision-making at the taluka level. Future research can extend this system across multiple districts and states to capture regional variability in climate, soil characteristics, and crop management practices. Such large-scale deployment would enhance the generalizability of the model and enable state-level or national-level yield forecasting [3], [9]..

Incorporating additional remote sensing sources, such as hyperspectral imagery and Synthetic Aperture Radar (SAR) data, can further improve prediction accuracy, particularly under cloud-covered conditions. The inclusion of soil moisture indices, groundwater availability, and pest or disease indicators would allow the system to model real-field conditions more comprehensively.

Advanced deep learning architectures, including Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks, may be explored to better capture spatial and temporal dependencies in satellite time-series data. Hybrid CNN–LSTM models could provide improved understanding of crop growth stages and seasonal variations.

Future enhancements may also involve the development of a farmer-centric mobile or web-based application that delivers early yield forecasts, crop health alerts, and advisory recommendations in regional languages. Integration with real-time IoT sensors deployed in agricultural fields could further enable continuous monitoring and adaptive prediction.

Collectively, these advancements would transform the proposed system into a scalable, intelligent, and climate-resilient agricultural decision support platform capable of improving productivity, reducing uncertainty, and promoting sustainable sugarcane farming practices [1], [6].

TABLE V
SEASON-WISE PERFORMANCE OF YIELD PREDICTION MODEL

Season	Samples	Mean Yield	MAE	RMSE	R^2
Adsali	3326	151.48	8.55	10.88	0.75
Pre-seasonal	3346	131.18	7.86	9.80	0.81
Suru	3328	109.40	6.64	8.35	0.84

CONCLUSION

This study presented an AI-driven framework for accurate sugarcane yield prediction by integrating multispectral satellite data, meteorological parameters, and agronomic information. The proposed system effectively addresses the limitations of traditional yield estimation methods such as manual Crop Cutting Experiments, which are time-consuming, labor-intensive, and lack spatial granularity.

By employing advanced machine learning models, particularly XGBoost and Random Forest, the system successfully captured complex non-linear relationships between vegetation health, weather variability, and crop management practices. Experimental results demonstrated strong predictive performance, with the proposed model achieving an R^2 value of up to **0.90**, indicating its capability to explain a significant portion of yield variability across different seasons and regions.

The feature importance and ablation studies further validated the significance of vegetation indices—especially NDVI—along with seasonal and irrigation-related agronomic factors in influencing sugarcane productivity. These findings highlight the importance of multi-source data fusion for reliable agricultural forecasting. The model also exhibited computational efficiency, making it suitable for large-scale and near real-time deployment.

Overall, this research demonstrates that combining remote sensing technologies with artificial intelligence provides a scalable, cost-effective, and data-driven solution for precision agriculture. The proposed framework has strong potential to support farmers, agricultural planners, and policymakers in improving crop monitoring, yield estimation, and resource management. This work lays a solid foundation for future intelligent agricultural decision-support systems aimed at enhancing productivity and sustainability in the sugarcane sector.

REFERENCES

[1] K. Narayansamy and I. Venkatachalam, "Predicting Sugar Yield from Sugarcane Using Machine Learning for Jaggery Production," *IEEE Access*, vol. 7, p. 1419, Apr. 2025.

[2] Y. Yuan, L. Yang, K. Chang, Y. Huang, H. Yang, and J. Wang, "DSCA-PSPNet: Dynamic Spatial-Channel Attention Pyramid Scene Parsing Network for Sugarcane Field Segmentation in Satellite Imagery," Jan. 2024.

[3] J. Som-ard, C. Atzberger, E. Izquierdo-Verdiguier, F. Vuolo, and M. Immitzer, "Evaluating Sugarcane Yield Estimation in Thailand Using Multi-Temporal Sentinel-2 and Landsat Data Together with Machine-Learning Algorithms," Sept. 2024.

[4] S. Kumar, M. Pant, and A. Nagar, "Forecasting the Sugarcane Yield Based on Meteorological Data through Ensemble Learning," *IEEE*, vol. 145, pp. 311–318, Oct. 2024.

[5] S. Akbarien, M. R. Jamnani, C. Xu, W. Wang, and S. Lim, "Plot-Level Sugarcane Yield Estimation by Machine Learning on Multispectral UAV Images," *KEAI Publishing*, Apr. 2023.

[6] S. Sridhara, S. B. R., and G. R. Kashyap, "Multistage Sugarcane Yield Prediction Using Machine Learning Algorithms," *Plants*, Feb. 2023.
[7] A. Das, M. Kumar, A. Kushwaha, R. Dave, and B. K. Bhattacharya, "Machine Learning Model Ensemble for Predicting Sugarcane Yield Through Synergy of Optical and SAR Remote Sensing," Mar. 2023.
[8] B. Alemán-Montes, "Models for the Estimation of Sugarcane Yield in Costa Rica with Field Data and Vegetation Indices," Jan. 2023.
[9] N. A. Patel, A. Danodia, and N. R. Patel, "Sugarcane Yield Forecasting Using Sentinel-2 Imagery and Machine Learning Algorithms," Apr. 2022.
[10] A. Das, M. Kumar, A. Kushwaha, R. Dave, K. K. Dakhore, K. Chaudhari, B. K. Bhattacharya, and S. Pokhariyal, "Field-Scale Assessment of Sugarcane for Mill-Level Production Forecasting Using Indian Satellite Data," Dec. 2021.
[11] R. Medar, V. S. Rajpurohit, and S. Medar, "Crop Yield Prediction Using Machine Learning Techniques on Multispectral Satellite Imagery," May 2021.
[12] F. H. S. Karp, "Sugarcane Yield Map Prediction Based on Satellite Imagery," *American Sugar Cane League Winter Meeting*, Feb. 2021.
[13] Y. Shendryk, R. Davy, and P. Thorburn, "Integrating Satellite Imagery and Environmental Data to Predict Field-Level Cane and Sugar Yields in Australia Using Machine Learning," Jan. 2021.
[14] Y. Li and L.-T. Yang, "Estimation of Sugarcane Yield Using a Machine Learning Approach Based on UAV-LiDAR Data," Aug. 2020.
[15] R. Medar, V. S. Rajpurohit, and A. M. Ambekar, "Sugarcane Crop Yield Forecasting Model Using Supervised Machine Learning," Aug. 2019.